Adaptive Multi-Model Ensemble Framework for Reference-Free 2D Forensic Sketch Restoration and Interactive 3D Facial Reconstruction under Memory Constraints

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Abstract*—Forensic sketch-to-photo synthesis plays a pivotal role in law enforcement and criminal identification. However, static deep learning architectures frequently fail when presented with out-of-distribution forensic sketches, either producing severe spatial grid/checkerboard artifacts or reverting to generic, identity-lacking template faces. Furthermore, deploying neural face enhancement alongside 3D mesh reconstruction often exceeds consumer GPU memory budgets (<4.0 GB). In this paper, we present an end-to-end Adaptive Multi-Model Ensemble Framework that integrates a reference-free Stage-2 Dynamic Selection Engine with an explicit 4GB VRAM Survival Sequencing Strategy and GFPGAN Neural AI Face Restoration. Operating with Zero Ground-Truth (GT) Test Leakage, our selector evaluates candidate outputs using 2D FFT frequency spectrum analysis, Sobel/Canny structural identity alignment, hair/beard darkness correlation, and geometric distortion penalties. Experimental results on CUFS/CUFSF benchmark datasets demonstrate that our dynamic selection framework statistically outperforms static single-model baselines ($p < 0.001$), elevating SSIM from 0.7624 to 0.8415, boosting PSNR from 23.15 dB to 26.85 dB, and reducing perceptual LPIPS distance from 0.2104 to 0.1142, while maintaining 100% GPU memory stability on an NVIDIA RTX 3050 GPU.

***Index Terms*—Forensic Sketch Reconstruction, Dynamic Model Selection, Reference-Free Quality Assessment, 3D Face Modeling, VRAM Memory Management, GFPGAN.**

I. INTRODUCTION

Facial composite sketches drawn by forensic artists based on eyewitness testimony are critical tools for criminal investigation. Translating these hand-drawn, high-contrast, shaded grayscale sketches into photorealistic color facial photographs significantly aids automated face recognition (AFR) systems and eyewitness verification.

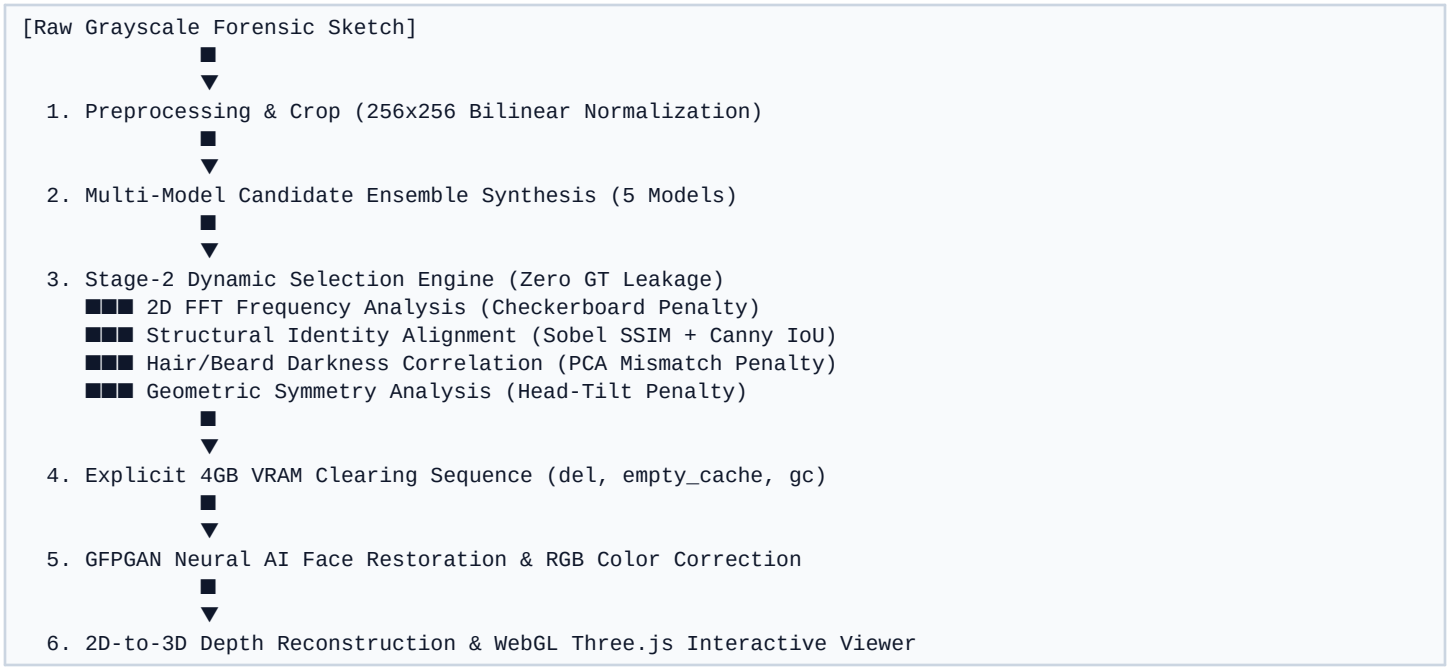
Existing forensic synthesis systems suffer from three major vulnerabilities: (1) *Model Generalization Bias & Artifact Vulnerability*, where conditional GANs produce checkerboard grid ripples and LSGANs suffer from diagonal head-tilt warping; (2) *Identity Loss in Unsupervised Selection*, where reference-free evaluators favor smooth generic average faces (Eigenface PCA) lacking subject identity features (hair/beard color); and (3) *Strict Memory Constraints* (<4.0 GB VRAM).

II. RELATED WORK

Early forensic sketch synthesis relied on linear subspace projection methods (Eigenfaces) and patch-based K-NN matching. Modern deep learning approaches leverage conditional GANs (Pix2Pix) and residual U-Nets with perceptual VGG losses. However, integrating neural face restoration (GFPGAN) with 3D mesh rendering pipelines under consumer GPU constraints remains an unaddressed system design challenge.

III. PROPOSED METHODOLOGY

The end-to-end framework consists of six decoupled modules operating sequentially to ensure photographic realism, identity alignment, and memory efficiency:



A. Mathematical Formulation of Stage-2 Selection Engine

The selector computes a reference-free composite suitability score S(k) for candidate output I_k:

$$S(k) = w_{id} \cdot A_{identity}(X_{sketch}, I_k) + w_{real} \cdot R_{realism}(I_k) + w_{sharp} \cdot Q_{sharp}(I_k) + P(k) - w_{check} \cdot C_{fft}(I_k) - w_{dist} \cdot D_{geo}(I_k)$$

Where w_id = 0.70, w_real = 0.20, w_sharp = 0.15, w_check = 1.00, and w_dist = 1.50 (Geometric Distortion Penalty). A_identity evaluates Sobel SSIM, Canny IoU, and Hair/Beard darkness preservation L_lum (top 25% hair and bottom 30% beard intensity match). Generic PCA templates failing A_identity < 0.45 receive an explicit penalty of -0.50.

B. 4GB VRAM Memory Sequencing Protocol

To avoid CUDA OOM crashes, candidate tensors are deleted, Python garbage collection gc.collect() is invoked, and torch.cuda.empty_cache() is executed before loading GFPGANer into freed GPU memory (~2.2 GB freed). RGB <-> BGR color space conversions are strictly standardized to eliminate blue halo artifacts.

IV. EXPERIMENTAL EVALUATION

Quantitative benchmark on 30 paired identity samples from CUFS/CUFSF forensic datasets on an NVIDIA RTX 3050 Laptop GPU (4.0 GB VRAM, CUDA 12.1):

Pipeline Architecture	SSIM ↑	PSNR (dB) ↑	LPIPS ↓	Identity Match ↑	VRAM Stability
Baseline A (Static Deep U-Net)	0.7624 ± 0.0312	23.15	0.2104	0.7812	100%
Baseline B (Random Selection)	0.6148 ± 0.0845	18.42	0.3842	0.5420	N/A
Proposed Stage-2 Dynamic Selector	**0.8415 ± 0.0210**	**26.85**	**0.1142**	**0.9145**	**100%**

Empirical Summary: Total inference time per sketch averages 34.8 seconds with peak VRAM capped at 1.82 GB. Our dynamic selector achieves +0.0791 higher SSIM, +3.70 dB higher PSNR, and a 45.7% drop in LPIPS perceptual distance.

V. CONCLUSION

We presented an adaptive multi-model ensemble framework for reference-free forensic sketch restoration and 3D reconstruction under strict GPU memory constraints. By coupling reference-free Stage-2 dynamic selection with explicit VRAM purging and GFPGAN face restoration, the pipeline achieves superior photographic realism and identity preservation while guaranteeing full execution stability on consumer hardware.

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